

Everything below is a *read* of one binary merge tree per jet. Clustering records, per step: **hist_p1,p2** (the two parent ids merged; p2=-1 beam merge = jet finished, -2 pad), **hist_child** (new id), **hist_d** (the distance below). Initial particles are ids 0..*n*-1; each merge makes a new id. E-scheme $\Rightarrow$ a pseudojet's 4-momentum is the *sum* of its constituents (why **pseudojet.p4** is just addition).

0. The clustering (foundation — anti-kt / kt / C-A)

Sequential recombination: repeatedly merge the pair with the smallest distance, or merge *i* with the beam.

$$d_{ij} = \min(k_{t,i}^{2p}, k_{t,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = k_{t,i}^{2p}, \quad \Delta R_{ij}^2 = \Delta y_{ij}^2 + \Delta \phi_{ij}^2.$$

The exponent *p* picks the algorithm: **kt** *p*=+1 (soft first), **Cambridge/Aachen** *p*=0 (purely angular, $d_{ij} = \Delta R^2/R^2$), **anti-kt** *p*=-1 (hard first, round jets). *y* = rapidity, *k_t* = transverse momentum. **hist_d** stores exactly this d_{ij} . [anti-kt 0802.1189 p2-3]

1. Exclusive jets — exclusive_jets_from_history (a kt concept)

Inclusive clustering runs to the end (every particle in some jet). **Exclusive** = stop early and read off the clusters that remain. Two stopping rules:

- *d_{cut}*: undo every pair-merge with $d_{ij} \geq d_{cut}$ (keep $d_{ij} < d_{cut}$). keep_pair = hist_d < d_cut
- *n_{jets}*: keep only the first $k = n_{init} - n_{jets}$ merges of the sequence, leaving exactly *n_{jets}* clusters. (For kt, *d* is monotonic $\Rightarrow$ “first *k* steps” = “smallest-*d* *k* merges” = FastJet’s undo-the-widest-*k*.)

Then each particle is assigned to the highest pseudojet it reaches through *kept* merges only (**_resolve_roots**). Anchor: *n_{jets}* = *n_{inclusive}* reproduces the inclusive partition. [FastJet manual p14, p20]

2. Grooming — groom_from_history (soft-drop / mass-drop)

Goal: strip soft wide-angle radiation, keep the hard 2-prong core (cleaner jet mass). **Walk each jet from its root down the harder branch**, undoing the widest split first. At each node with parents *i, j* (let $z = \frac{\min(p_{t,i}, p_{t,j})}{p_{t,i} + p_{t,j}}$, $\Delta R = \Delta R_{ij}$), test the **soft-drop condition**

$$z > z_{cut} \left(\frac{\Delta R}{R} \right)^\beta.$$

- **passes** $\rightarrow$ this node is the groomed jet; stop.
- **fails** $\rightarrow$ drop the softer parent, recurse into the harder one.
- reach a single particle without passing $\rightarrow$ *untagged*.

Special cases: $\beta=0$ is the **modified Mass-Drop Tagger (mMDT)** (condition becomes just $z > z_{cut}$). Passing also $\mu=$ adds the **original mass-drop** requirement $\max(m_i, m_j) < \mu m_{node}$ (MDT default $\mu=0.67$, $y_{cut}=0.09$). The walk depth is the $\mathcal{O}(\log_2 n)$ tree descent. [BDRS 0802.2470; mMDT 1307.0007 p24-26; SoftDrop 1402.2657 p3,p6]

3. Lund coordinates — lund_coordinates_from_history

De-cluster each jet (widest split first) and, at every split, emit the kinematics of that emission — the inputs to the **primary Lund plane** (a standard substructure / ML representation). Per split, from the two parents *i, j*:

$$z = \frac{\min(p_{t,i}, p_{t,j})}{p_{t,i} + p_{t,j}}, \quad \Delta R = \Delta R_{ij}, \quad k_t = \min(p_{t,i}, p_{t,j}) \Delta R,$$

and the code stores the 6-tuple $(z, \Delta R, k_t, \ln \frac{1}{\Delta R}, \ln k_t, d)$; the Lund plane is the $(\ln \frac{1}{\Delta R}, \ln k_t)$ scatter of a jet’s splits. The last channel *d* equals **splitting_scales** exactly (the built-in cross-check). Bounds on real splits: $z \in (0, \frac{1}{2}]$, $\Delta R \leq R$. [Lund plane 1807.04758 p4-6]

Mental model: clustering builds a binary tree per jet. **F1** cuts the tree at a chosen depth (exclusive jets); **F2** walks it top-down peeling soft branches (grooming); **F3** logs the physics of every split (Lund). The pages that follow are the primary-source definitions of each formula above.